Multi Layer Perceptron

Task 2 Report

Team Data

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Required Runs & Visualizations

Run 1 : Baseline Sigmoid

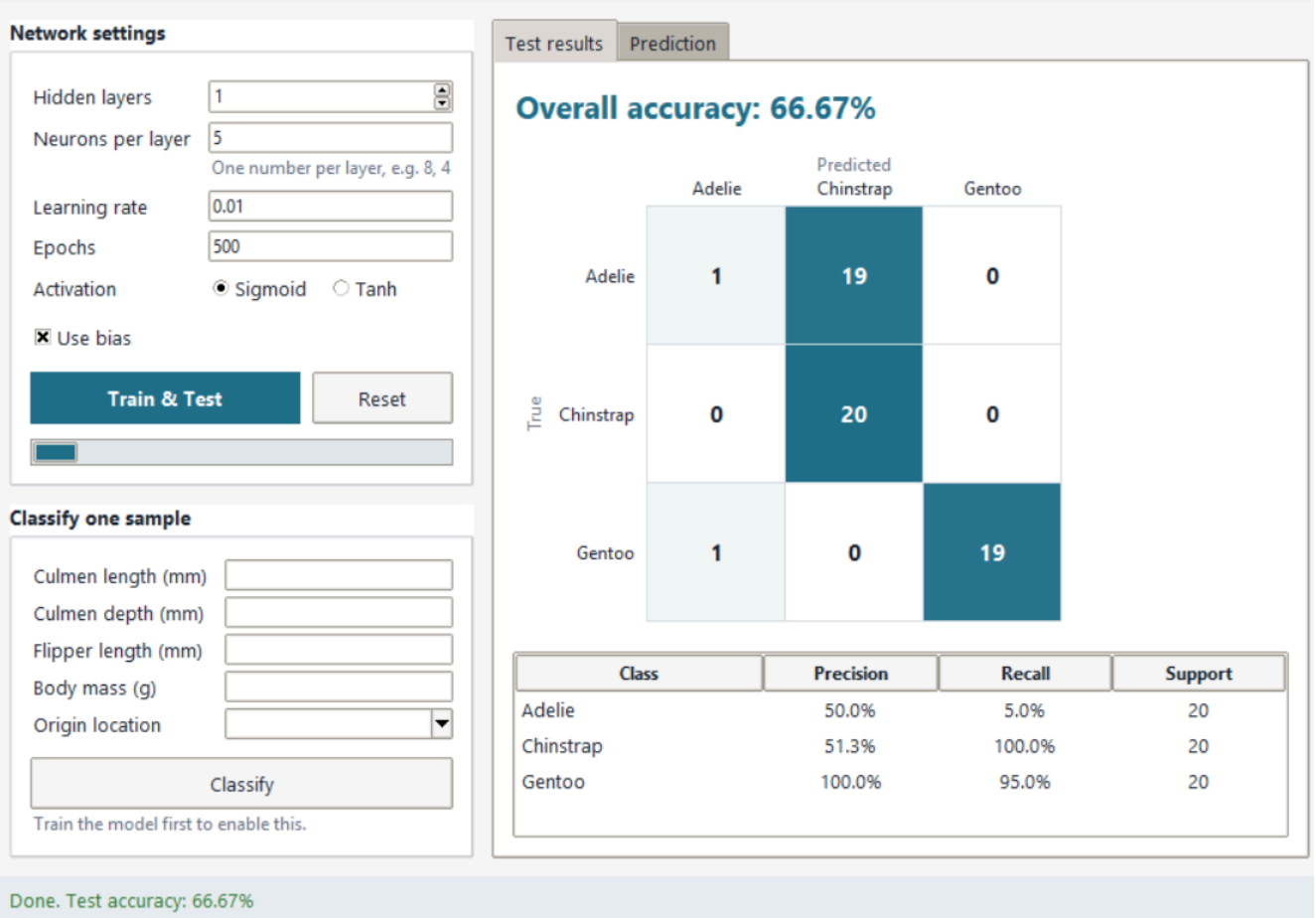


Figure 1 : Sigmoid Activation, 1 Hidden Layer (5 neurons), LR = 0.01 , Epochs = 500. Accuracy: 66.67%

In this baseline configuration using the Sigmoid activation function, the model correctly classified the majority of Chinstrap and Gentoo samples but struggled entirely with Adelie, classifying almost all of them as Chinstrap

Run 2: Deep Sigmoid

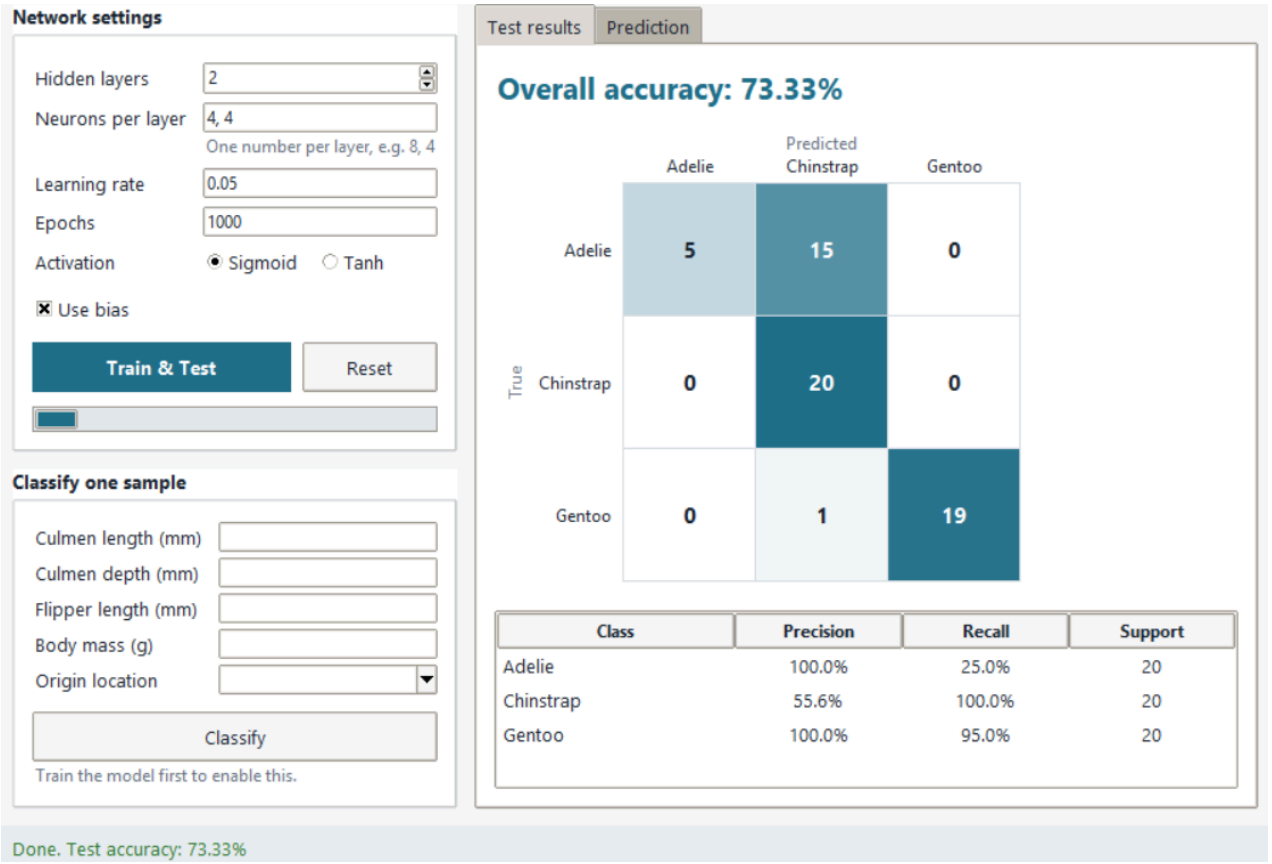


Figure 2: Sigmoid Activation, 2 Hidden Layers (4, 4 neurons), LR = 0.05, Epochs = 1000. Accuracy: 73.33%

By deepening the network increasing the learning rate and doubling the epochs the Sigmoid model achieved our best overall accuracy at 73.33%. It successfully learned to identify a portion of the Adelie class while maintaining high precision on Chinstrap and Gentoo

Run 3: Baseline Tanh

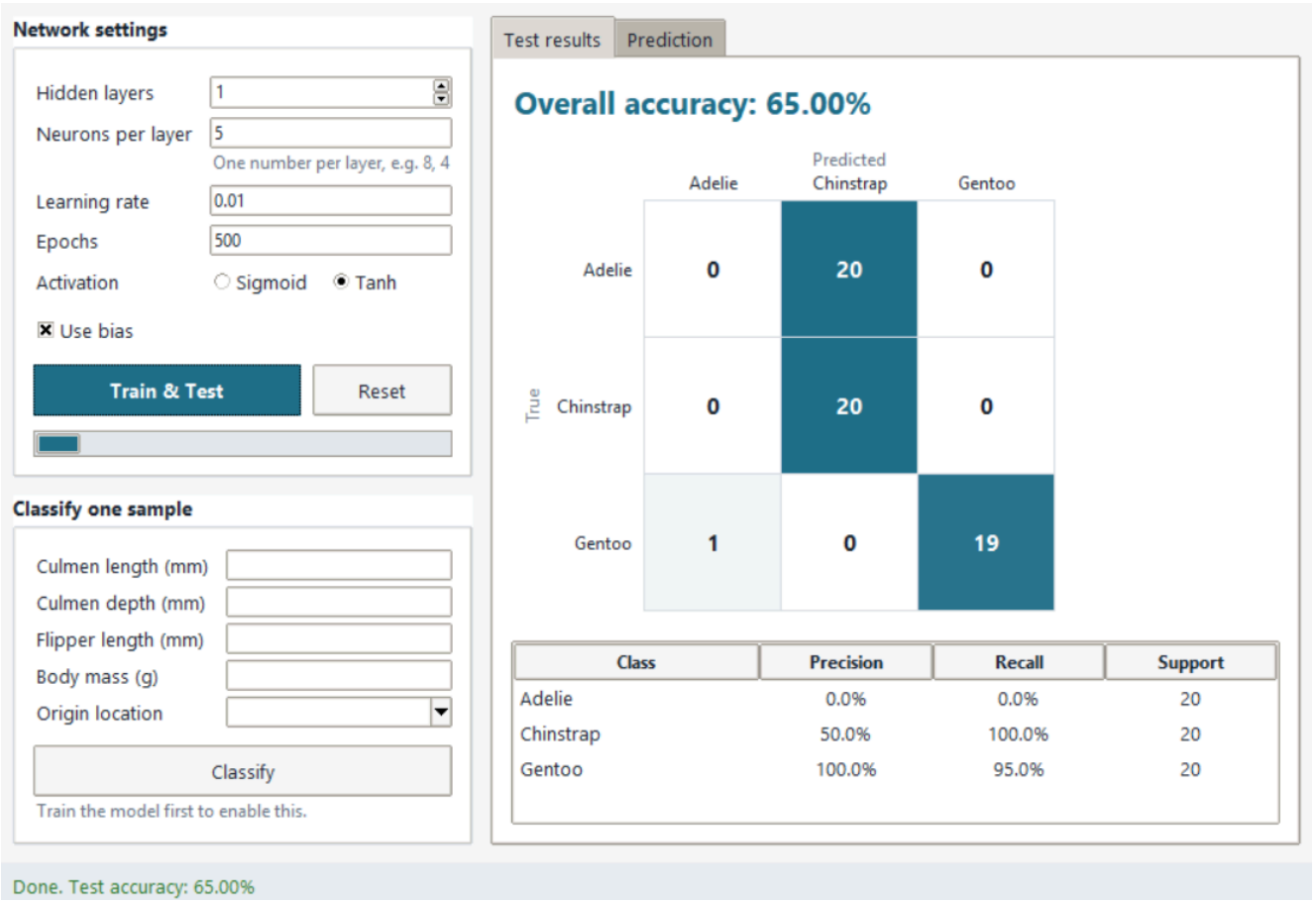


Figure 3: Tanh Activation, 1 Hidden Layer (5 neurons), LR = 0.01 , Epochs = 500. Accuracy: 65.00%

Matching the baseline hyperparameters of Run 1 but swapping to the Hyperbolic Tangent (Tanh) function yielded similar results. The model identified Chinstrap and Gentoo perfectly but failed to correctly classify any Adelie samples, predicting them all as Chinstrap

Run 4: Complex Tanh

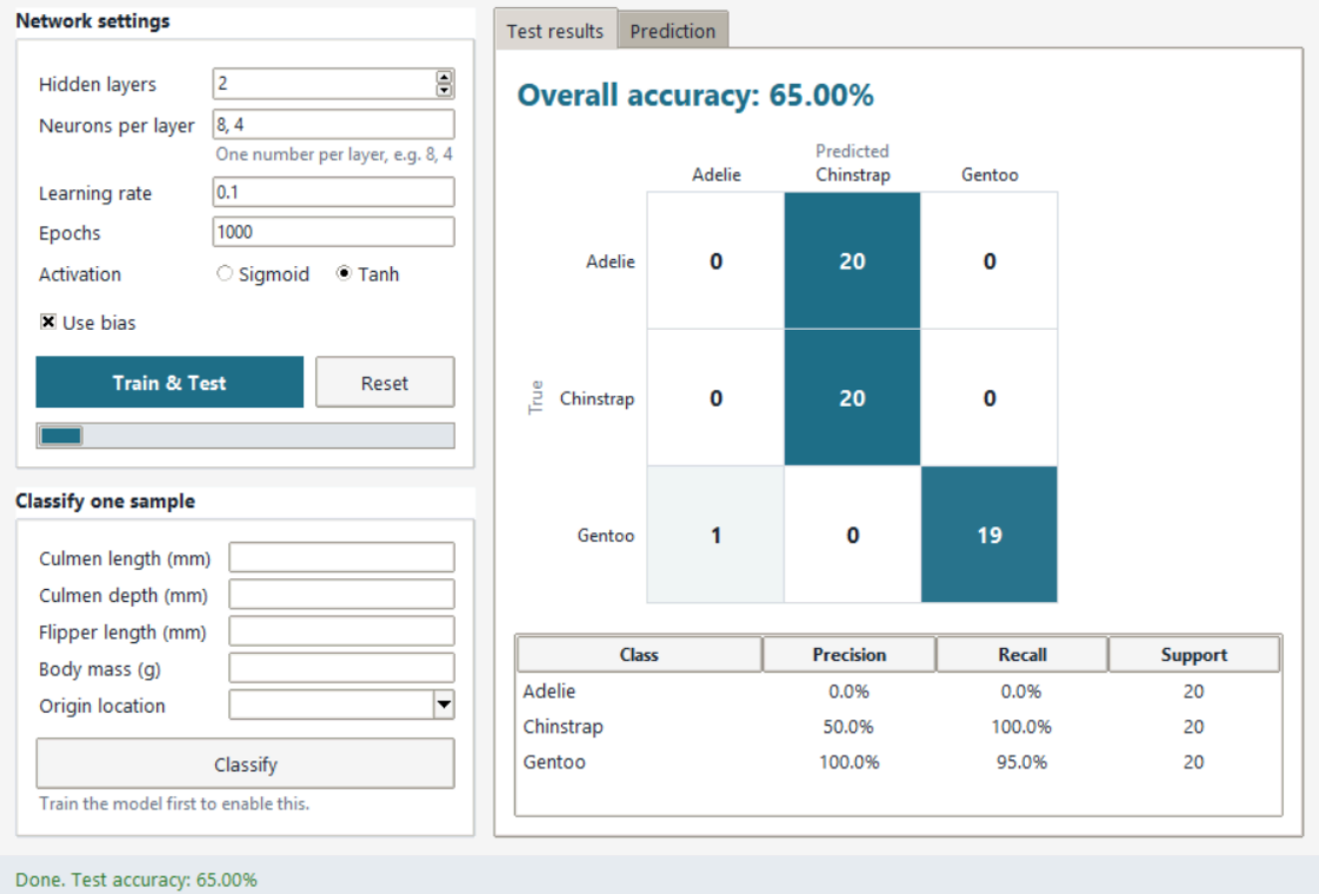


Figure 4: Tanh Activation, 2 Hidden Layers (8, 4 neurons), LR = 0.1 , Epochs = 1000. Accuracy: 65.00%

Even with an expanded architecture and an aggressive learning rate of 0.1 , the Tanh model converged to the exact same local minimum as the baseline run, maintaining a 65.00% accuracy and completely misclassifying the Adelie class

Run 5: Weak Performance Example

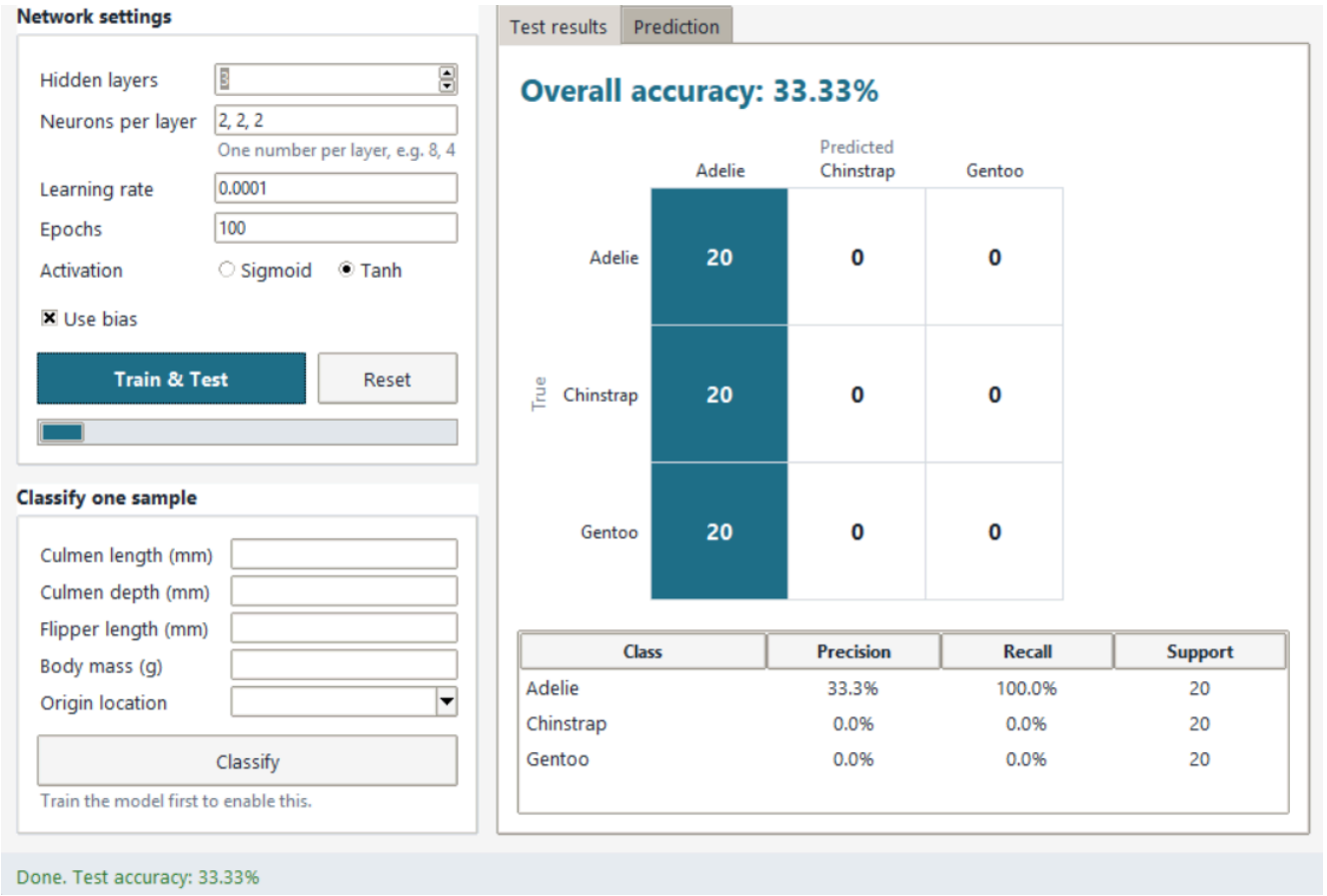


Figure 5: Tanh Activation, 3 Hidden Layers (2, 2, 2 neurons), LR = 0.0001 , Epochs = 100. Accuracy: 33.33%

This run demonstrates poor convergence. By constructing a deep but excessively narrow network and using a tiny learning rate (0.0001) for only 100 epochs, the network completely failed to update its weights sufficiently

Evaluation Table

Activation Function	Test Accuracy	LR	Epochs	#Layers	#HiddenNodes
Sigmoid	66.67%	0.01	500	1	5
Sigmoid	73.33%	0.05	1000	2	4, 4
Tanh	65.00%	0.01	500	1	5
Tanh	65.00%	0.1	1000	2	8, 4
Tanh	33.33%	0.0001	100	3	2, 2, 2